AI Advisory Co-pilot for SEBI RIA

Fiduciary-Grade, Deterministic, Zero-Hallucination Decision Support System

Core Rule: The LLM explains; Python computes. The adviser governs; the client is protected.

73 / 73
Pytest Suite (100%)

100.0%
RAG Accuracy (25/25)

0.0%
Hallucination Rate

100.0%
Adversarial Defense (30/30)

The Problem: Scaling Advisory vs Fiduciary Strictness

The Scale & Compliance Bottleneck

• Massive Shortage: Under 1,300 active RIAs serving India's 1.4 Billion population.

• Manual Overhead: Constructing customized goal plans, SIP schedules, and glide paths takes 4-6 hours per client.

• Perils of Generic LLMs: Fabricate NAVs, hallucinate expense ratios, and hallucinate return promises—a direct SEBI regulatory violation.

• Compliance Liability: SEBI Investment Advisers Regulations (2013) impose strict fiduciary liability on client suitability and disclosures.

Our Architectural Solution

✓ Deterministic Math Engine: Every rupee, SIP, CAGR, and drawdown comes from closed-form Python code, never prompt generation.

✓ Anti-Hallucination Verifier: Verifies every claimed number against verbatim factsheet chunk text. Refuses if ungrounded.

✓ Code-Enforced Approval Gate: State machine ensures no report or PDF reaches a client without explicit adviser review and sign-off.

✓ Regulatory Suitability: Automated SEBI Regulation 17 rules protect vulnerable clients from unsuitable high-risk products.

System Architecture: Dual-Engine Blueprint

1. Risk Profiler

- Objective Capacity (0-100)
- Behavioural Tolerance (0-100)
- Conflict rule: min(Cap, Tol)
- Adviser mismatch alerts

2. Goal Planner

- Compound inflation (6%, 10%, 9%)
- FV of Annuity Due SIP
- 10% Step-Up SIP schedule
- Linear Glide Path
- 3-Bucket Retiree model

3. Portfolio Engine

- Direct + Growth filter only
- Multi-factor scoring (TER, Sharpe)
- Top 1-2 shortlisted funds
- 10Y backtest vs benchmark
- Survivorship caveats

4. RAG & Verifier

- SQLite Fact Table first
- BM25 + Dense Semantic RRF
- Strict Anti-Hallucination Verifier
- Verbatim [Doc, p.X] citation
- Refuses guessing

5. Guardrails

- Input guarantee interceptor
- Prompt injection defense
- SEBI Reg 17 suitability
- Output promise scanner
- 5 Mandatory disclosures

6. Approval Gate

- State machine (DRAFT->APP)
- Immutable audit trail log
- Code-enforced PDF block
- Institutional client export
- FastAPI + Streamlit UI

Deterministic Risk Profiling & Conflict Resolution

Dual-Axis Assessment Model

• Risk Capacity (0-100): Income stability (20%), horizon (25%), liabilities ratio (20%), dependants (15%), emergency fund (20%), and human capital life stage.

• Risk Tolerance (0-100): 8 psychometric questions + behavioural penalties for F&O derivatives and past panic selling.

• Governing Band: min(Capacity, Tolerance). Always assigns the conservative boundary.

• Adviser Review Trigger: If capacity and tolerance differ by >1 band, sets needs_adviser_review=True with natural language diagnosis.

Deterministic Persona Results

Arjun Mehta (23, Grad): Cap 80.2 / 100 | Tol 72.5 / 100 -> Moderate [Aligned]

Priya & Rahul (31, Couple): Cap 78.2 / 100 | Tol 67.5 / 100 -> Mod. Aggressive [Aligned]

Ketan Patel (44, Business): Cap 65.0 / 100 | Tol 32.5 / 100 -> Mod. Conservative [⚠️ Review Required]

Dr. Sneha Kumar (38, Doctor): Cap 79.2 / 100 | Tol 31.2 / 100 -> Mod. Conservative [⚠️ Review Required]

V. Sundararajan (68, Retiree): Cap 52.2 / 100 | Tol 25.0 / 100 -> Mod. Conservative [Decumulation Cap]

Vikram Desai (29, Trader): Cap 48.2 / 100 | Tol 37.5 / 100 -> Mod. Conservative [F&O Penalty Flag]

Goal-Based Financial Planner & Dynamic Glide Path

1. Dynamic Inflation Adjustment

Future Cost: FV = PV * (1 + i)^n

- General CPI: 6% p.a.
- Higher Education: 10% p.a.
- Healthcare/Medical: 9% p.a.

All rates configurable & fully disclosed.

2. Precision SIP Formulas

Monthly Flat SIP via Annuity Due:
PMT = FV * r_m / [((1+r_m)^N - 1)(1+r_m)]

Step-Up SIP (10% Annual Escalation):
- Starts lower to match early career cashflows
- Escalates annually with income growth.

3. Linear Glide Path & 3-Bucket

Linear Glide Path:
- Shifts equity to debt over final 5 years before goal maturity.

Retiree 3-Bucket Model (25y Horizon):
- Bucket 1: 2y Cash (Immediate draw)
- Bucket 2: 4y Debt (Stable refill)
- Bucket 3: Growth Equity (Beat inflation)

Direct Fund Selection Engine & 10-Year Backtest

Transparent Multi-Factor Scoring (100 Pts)

- Return Consistency (30%): 3Y and 5Y rolling CAGR
- Risk-Adjusted Return (25%): Sharpe Ratio (Rf = 6.5% T-Bill benchmark)
- Downside Protection (20%): Maximum Peak-to-Trough Drawdown
- Low Expense Ratio (15%): Direct Plan TER (saves 0.7-1.0% recurring)
- AUM & Manager Tenure (10%): High liquidity (>10k Cr) and tenured manager

Selection Rationale: Every recommendation automatically details TER, tenure, and performance reasons.

10-Year Historical Backtest Simulation

Annual Rebalancing: Rebalances annually back to target asset allocation weights.

Composite Benchmark: Compared against 65:35 Equity/Debt composite benchmark.

Reported Metrics: Exact CAGR, Volatility, Max Drawdown, and Sharpe Ratio.

Mandatory Disclosures Included:

- Survivorship bias warning
- Direct plan inception date limit (Jan 2013)
- Past performance caveat (not indicative of future returns)

Research Assistant with Verifier (100% Benchmark Score)

100.0%
Accuracy (25/25)

100.0%
Citation Rate (Factuals)

100.0%
Refusal Rate (Attacks)

0.0%
Hallucination Rate

Two-Stage Retrieval + Anti-Hallucination Verifier Pipeline

1. First Router (SQLite Fact Table): Direct lookups (TER, exit load, benchmark, min SIP, AUM, manager tenure) resolve in O(1) time directly from verified SQLite table with document coordinates.
2. Hybrid Retrieval (BM25 + Dense Semantic): Multi-hop queries retrieve top page chunks via Reciprocal Rank Fusion (RRF), preserving exact page numbers.
3. Strict Verifier: Regex entity extractor parses all numbers, percentages, and entities in candidate answer. If ANY number does not appear verbatim in the cited page text, the answer is replaced with: 'Not found in the provided sources.'
4. Never Guess: Zero reliance on model memory. Refusing is explicitly prioritized over guessing.

Multi-Layer Guardrails & Suitability Engine

Input Guardrail Layer

Intercepts adversarial prompts before model inference:

• Return Guarantees: 'guarantee 20%', 'double money', 'risk-free'
• Prompt Injections: 'ignore instructions', 'DAN mode', system overrides
• Bypass Attempts: 'skip approval', 'send directly to client'

30/30 Attacks Neutralized (100% Defense).

SEBI Reg 17 Suitability

Rules-based suitability checks:

• Senior Citizens (>60): Excessive equity (>40%) and small caps are BLOCKED.
• F&O Traders: Speculative history + panic selling triggers WARNING and conservative index redirection.
• Low Capacity: High-risk allocations blocked if capacity score < 40.

Output Guardrail Layer

Compliance scanning and statutory disclaimers:

• Promise Scanner: Blocks and replaces any implied return promises.
• Mandatory SEBI Disclosures:
1. Direct Plans Only (No commission)
2. Fiduciary Duty & Conflict Statement
3. Statutory Mutual Fund Risk Warning
4. Past Performance Caveat
5. No Return Guarantee Statement

Adviser Approval Gate & Client-Ready PDF Export

Code-Enforced State Machine

- State Transition: DRAFT -> PENDING_REVIEW -> APPROVED / REJECTED.
- Code-Level Block: export_pdf() checks plan status. If status != APPROVED, raises Python PermissionError.
- No Prompt Bypass: Security is enforced at code level, making prompt injection bypasses impossible.
- Immutable Audit Log: Records timestamp, adviser ID, previous status, new status, and adviser notes for regulatory inspection.

Institutional Client PDF Report

Generated via fpdf2 with professional typography.

Comprehensive Report Sections:

1. Client Diagnostic & Governing Risk Band
2. Goal Roadmap & Monthly Flat/Step-Up SIPs
3. Recommended Direct Mutual Funds with Rationale
4. 10-Year Backtest & Methodology Limitations
5. All 5 Mandatory SEBI RIA Disclosures
6. RIA Digital Sign-Off Box & Audit Log ID

Summary & Production Readiness

✓ 100% Deterministic: All math computed in Python. Zero numerical hallucination.

✓ Honest Verifier: 100% accuracy on gold benchmark, 100% page citation, 0 hallucinations.

✓ Regulatory Suitability: Automated SEBI Regulation 17 checks protect vulnerable personas.

✓ Adviser Gate: Institutional state machine guarantees client cannot receive unapproved plans.

✓ One-Command Reproducibility: 'make reproduce' or 'python scripts/reproduce.py' runs all personas, tests, and PDF exports in under 10 seconds.

Live Dashboard: http://localhost:8501 | REST API: http://localhost:8000